

Kinetic Delivery of Whole Blood via Autonomous Drones: Reducing Casualties for Air-Parity Scenarios

Abstract

Brief summary of paper. (Write this after everything else)

Introduction

Introduction to paper. Introduce all the major themes that will be explored.

Examining Modern Battlefield Medicine

Briefly explain how dangerous a battlefield is and what kinds of injuries are possible. Discuss how this is different from past conflicts.

Medevac During the War on Terror

Explain a typical medevac procedure during the war on terror.

Lessons from Ukraine: Consequences of Air Parity

Discuss how air-parity makes air-based medevac impossible (when enemy air defense is present), and what effect this has on casualty rates.

Limitations of Existing Remedies

Talk about two approaches to this problem: blood thermoses and “walking blood banks”. Explain the drawbacks to these approaches.

Sustaining the Fight through Autonomous Blood Logistics

Present the idea of using drones for delivery. Describe the moving refrigerator trucks which can deploy these drones behind the main front of a battle space, as well as other relevant infrastructure.

Advantages of Drones as a Delivery Platform

Discuss why drones can overcome the problems of the previous two remedies. In particular, go into how they are too small for most countermeasures, and how they are agile enough to get close for an accurate delivery maneuver.

Delivery Under Duress

Briefly explain why typical drone delivery techniques are not suitable for blood delivery, and why kinetic delivery is necessary.

The Sanguine Soldier

Discuss the idea of the “Sanguine Soldier”, a theoretical concept in which a soldier who should have died from their wounds is able to continue fulfilling some level of combat capacity due to a constant supply of transfused whole-blood that keeps them alive.

Proof of Concept: AeroMed Project Gamma

Present the Project Gamma prototype. Describe it’s purpose of demonstrating the basic premises described thus far.

Autonomous Conversion Kit for Repurposing Contaminated Technology

Explain how an autonomous conversion kit is used so that an otherwise contaminated drone can still be deployed.

Target Condition Satisfiability with Hitpolys

Describe a hitpoly and how it’s used for targeting algorithms.

Experimental Demonstration

Describe the set up of the experiment, and how a package will be thrown through a target hoop/window.

Metrics

Come up with some measurable metrics to show the efficacy of our approach.

Results

Run the experiment and analyze the resulting metrics.

Conclusion

Briefly describe all the topics, concepts, and ideas this paper has visited. Make an assertion about what our work means, and the impact it might have.